

Kaveri Raut

7220813813 (rautkaveri88@gmail.com) | [Portfolio](#)

AI/ML Engineer with 2+ years of experience developing machine learning and deep learning solutions using Python. Hands-on experience with NLP, neural networks, Transformers, BERT, LLMs, RAG, LangChain, and vector databases. Experienced in building AI-powered applications, developing inference APIs, and deploying scalable ML solutions on Stramlit.

PROFESSIONAL EXPERIENCE

GenAI Engineer- Inphynous Technology Solution | June 2024 – Present

JEE Question Generator – AI-Powered Question Generation Platform

Python, LangChain, HuggingFace Embeddings, ChromaDB, RAG, PyMuPDF, LLMs

- Built an AI-powered question generation system using **JEE Main previous-year papers and textbook content** to generate topic-specific questions based on historical question patterns.
- Developed a **document processing and RAG pipeline** using PyMuPDF for PDF extraction, text chunking, HuggingFace embeddings, and ChromaDB for semantic retrieval.
- Implemented **vector-based similarity search** to retrieve relevant textbook and previous-year question context before generating questions with an LLM.
- Designed topic/chapter mapping to connect textbook knowledge and historical questions, enabling **context-aware question generation and question probability analysis**.

TECHNICAL SKILLS

Generative AI & LLMs:	NLP, Transformers, BERT, LLMs, RAG, LangChain, Prompt Engineering, Vector Databases, FAISS, ChromaDB.
Machine Learning:	Regression, Classification, Clustering, Feature Engineering, Model Evaluation
Programming:	Python, SQL, Javascript.
Libraries:	NumPy, Pandas, Scikit-learn, TensorFlow, Keras.
Cloud & Deployment:	AWS (EC2, S3, SageMaker), Flask APIs.
Statistics:	ANN, CNN, RNN, LSTM, GRU, Tensorflow, Keras.
Databases:	MySQL, Postgress, Supabase.

PROJECTS

DevCook – AI Website Generation Platform- [VISIT](#)

LLMs, Prompt Engineering, React, Next.js, Python/Node.js, REST APIs

- Built an AI-powered platform that converts natural-language user prompts into functional React-based websites using LLM-driven prompt-to-code generation.
- Designed prompt-processing and code-generation workflows to improve the **accuracy and consistency of generated application code**.
- Developed scalable APIs for AI processing with **real-time website preview and export functionality**, enabling users to iterate on generated applications.
- Integrated frontend and backend components to create an end-to-end **AI application generation workflow** from user prompt to executable UI.

Next Word Prediction — NLP / Deep Learning

Python, NLP, TensorFlow, Keras, Tokenization, Word Embeddings, RNN, LSTM, Deep Learning

- Developed an NLP-based **next-word prediction model** using sequential text data to predict the most probable next word given preceding context.
 - Implemented **text preprocessing, tokenization, vocabulary construction, and sequence generation** to prepare textual data for model training.
 - Built a deep learning architecture using **word embeddings and recurrent neural networks** to capture sequential dependencies in language.
 - Trained and evaluated the model for **context-aware next-word prediction**, analyzing model performance across different input sequences.
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EDUCATION

B.Tech – Computer Science & Engineering

CSMSS Chh. Shahu College of Engineering, Aurangabad

2024 | 70%